

1. What is Statistics?

Statistics is the science of collecting, organising, analysing and interpreting data.

Descriptive vs inferential

Descriptive statistics summarise data you have (mean, median, charts).

Inferential statistics draw conclusions about a population from a sample.

Population and sample

Population: the entire group of interest. Its measures are parameters (μ , σ).

Sample: the subset we observe. Its measures are statistics ($\bar{x}$, s).

2. Data Types

Scales of measurement

Nominal: categories with no order (blood group, city).

Ordinal: ordered categories (grades A, B, C).

Interval: equal spacing, no true zero (temperature in Celsius).

Ratio: equal spacing with a true zero (height, income).

3. Descriptive Statistics

Central tendency

Mean = sum of values / n. Median = middle value. Mode = most frequent value.

The median is robust to outliers; the mean is not.

Dispersion

Range = max - min. Variance = average squared deviation from the mean.

Standard deviation = square root of variance.

IQR = $Q3 - Q1$, the spread of the middle 50% of the data.

Example

Values 4, 8, 6, 2: mean = $20 / 4 = 5$, median = $(4 + 6) / 2 = 5$.